PFC official linear restitution

Complete Md=0 row: 16 original balls | Automatic damping | 29 October 2026

PASS	All 64 parameter/run checks meet the published ABSOLUTE coefficient-error target <0.01.

Sixteen normal damping ratios

beta	Contact e0	c_n (N s/m)	CPU e_h dt	CPU e_h dt/2	Abs. error dt	Abs. error dt/2
0	1.00000000	0	1.00000133	0.99999920	0.00000133	0.00000080
0.01	0.96907090	1.862483	0.96887616	0.96887555	0.00019475	0.00019536
0.02	0.93908956	3.724965	0.93872360	0.93872728	0.00036596	0.00036228
0.03	0.91001862	5.587448	0.90952129	0.90953161	0.00049733	0.00048701
0.04	0.88182262	7.449931	0.88086808	0.88107675	0.00095454	0.00074586
0.05	0.85446789	9.312414	0.85342489	0.85368077	0.00104301	0.00078712
0.1	0.72924761	18.62483	0.72754258	0.72763453	0.00170503	0.00161308
0.2	0.52662060	37.24965	0.52401744	0.52372422	0.00260316	0.00289638
0.3	0.37232610	55.87448	0.36830393	0.36873323	0.00402218	0.00359287
0.4	0.25382672	74.49931	0.24887051	0.24939242	0.00495621	0.00443430
0.5	0.16303353	93.12414	0.15730785	0.15782878	0.00572568	0.00520475
0.6	0.09478022	111.749	0.08815474	0.08886252	0.00662548	0.00591771
0.7	0.04598791	130.3738	0.03896948	0.03936429	0.00701843	0.00662362
0.8	0.01516462	148.9986	0.00689825	0.00716984	0.00826637	0.00799478
0.9	0.00152376	167.6234	0.00000000	0.00000000	0.00152376	0.00152376
1	0.00000000	186.2483	0.00000000	0.00000000	0.00000000	0.00000000

e_h=sqrt(max(z_first_apex-R,0)/(1-R)). The source computes absolute differences and publishes a fixed-step maximum below 0.01; it has no automated assertion. CPU/GPU results are audited separately for every parameter.

Unchanged source inputs and exact contact mapping

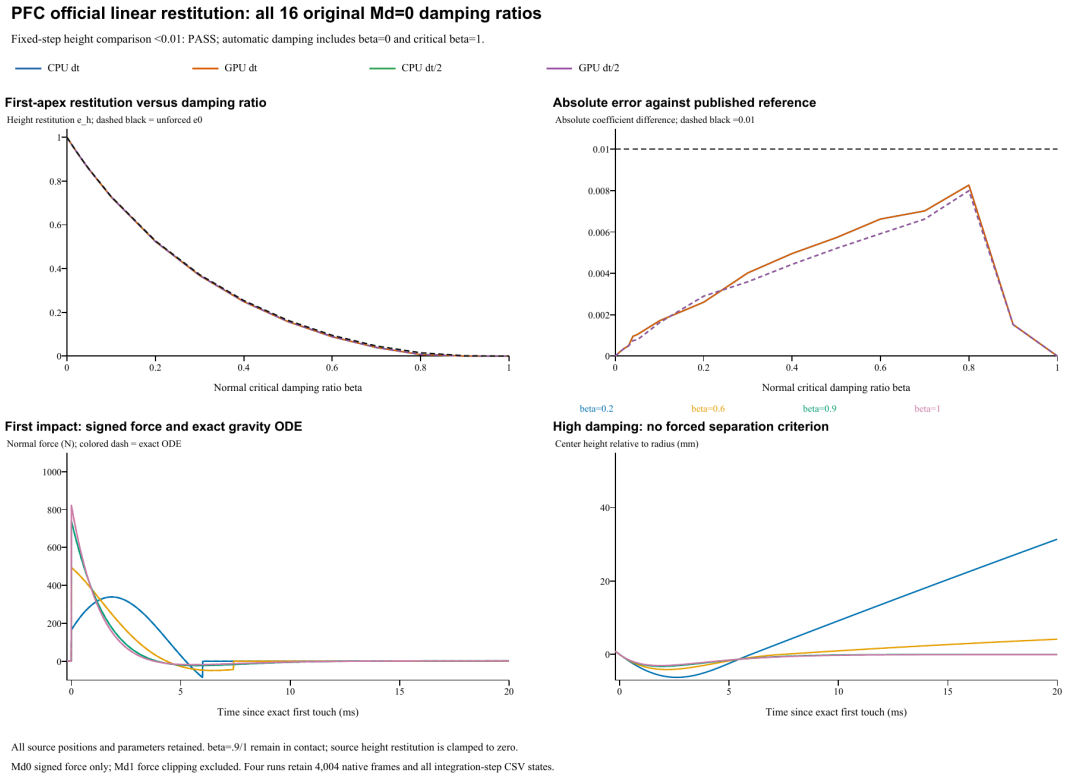
R=0.025 m; density 2650 kg/m^3; initial height 1 m; kn=50000 N/m; ks/friction=0; gravity 10 m/s^2 downward; zero initial velocities/spins. All 16 original positions are retained. Four CPU/GPU runs use dt=10/5 us to 1 s; native frame cadence 1 ms. The Md1 no-tension row is excluded.

Common floor e=1 and sphere e=e_contact/(2-e_contact) invert the native harmonic mixing exactly. The source specifies contact beta; no coefficient is fitted. All 153 pairs use automatic damping and zero manual overrides. Beta=0 gives c_n=0; beta1/e_contact=0 gives automatic critical damping.

Source identity and qualification

PFC 7.0 official verification | Fresh source hashes.
PFC is proprietary Itasca software with public examples. Local PFC was not executed. This reproduces the complete 16-ball Md0 row, not the full 32-ball two-mode example.

PFC Md0: native histories and checks



All four native runs and independent mechanics

Run	Max abs. e error	Force residual (N)	Impulse residual	Target
CPU dt	0.00826637	5.7e-14	1.1e-12	PASS
GPU dt	0.00826637	1.1e-13	1.1e-12	PASS
CPU dt/2	0.00799478	5.7e-14	4.3e-12	PASS
GPU dt/2	0.00799478	1.1e-13	4.3e-12	PASS

Row maximum absolute coefficient error falls from 0.00826637 to 0.00799478. Individual errors are not uniformly monotonic: beta=0.2 changes from 0.00260316 to 0.00289638. Contact onset and the sampled first-apex estimator depend on dt.

Gravity stays active during contact. At beta=0.8, unforced e0=0.01516462, exact gravity-aware e_h=0.00741005; relative errors to e0 are 54.51% / 52.72%. These do not replace the published ABSOLUTE <0.01 target. Beta=0.9/1 remain in contact, with e_h=0; a separation velocity is not forced.

CPU/GPU maximum whole-history z / vz / normal-force differences are 1.18e-14 m / 1.42e-12 m/s / 2.05e-10 N. Half-kick reconstruction includes gravity; raw saved force excludes it.

Each contact verifies the automatic dashpot, overlap geometry and spring energy; all numerical drifts/kicks and gravity-plus-floor impulses are audited. Beta>=0.6 passes supplemental late-time equilibrium checks. Signed Md0 unloading is preserved; Md1 force clipping is not mapped.

Complete retained results and verified scope

All 4004 native frames, 600004 every-step wide CSV rows (9,600,064 sphere states), 2,080,520 active-contact rows, four loadable projects and backend logs are retained. Other transverse/spin/torque channels remain inactive and are checked at every step. Frame cadence is 1 ms; numerical dt is 10/5 us.

Run manifest | All 64 comparisons | Independent audit | Complete native ZIP | Frame/hash audit

Real SimulationSession runs use a stable helper executable statically linked to the current native CPU/CUDA libraries; historical GUI binary identity is not asserted. Scope: the complete Md0 normal-contact row, including zero and critical damping. Active friction and other PFC models require separate coverage.